

Adam Mrotek

Full-Stack & AI Engineer

London, UK (Hybrid / Remote) adam.mrotek@gmail.com [linkedIn](#) [gitHub](#) [portfolio](#)

Full-stack engineer with 4 years of experience specialising in type-safe UI architectures (TypeScript, Vue 3, React) and robust backend services (Python, FastAPI). Actively focused on applied LLM product engineering, building production-ready autonomous tooling, and owning features from technical roadmap to containerized delivery.

EXPERIENCE

Front-end Engineer — Croud

Jul 2023 – Mar 2026

London, UK (Hybrid)

- **Core Platform:** Owned end-to-end feature delivery across marketing suite. Migrated legacy Nuxt surfaces to clean, type-safe Vue 3 / TypeScript patterns.
- **Social Media Influencer Platform:** Contributed front-end features and supporting FastAPI endpoints to an influencer-marketing platform built in-house by a 5-engineer team — delivering against a multi-million-pound external build quote and matching strict Figma specs.
- **Taxonomy Tool:** Built the front-end (plus supporting backend tweaks) for an internal app aligning marketing tags across teams — projected to save ~£100k/month (pre-build estimate) by removing tagging inconsistency.
- **AI Tooling:** Built the front-end applications used to present internal AI tools and agentic pipelines in client pitches, plus a custom Claude sub-agent skill that converts Figma designs directly into our codebase — driving cross-team adoption and supporting upsell opportunities.

Stack: Vue 3, TypeScript, FastAPI, Nuxt, Tailwind, Shadcn, Docker, Claude/MCP, Vitest, Playwright, Storybook, Figma

Customer Engineer — Hullabalook

Apr 2022 – Jul 2023

London, UK

- **Discovery Applications:** Shipped multiple reactive retail discovery apps using React and Vue/Nuxt environments over high-frequency CI/CD pipelines.
- **Canvas Architecture:** Programmed canvas-driven visual mapping systems and interactive layouts utilising Fabric.js and the native HTML5 Canvas API.
- **Data Pipelines:** Standardised unstructured dataset normalisation via Google Cloud Platform (GCP) and Python scripts, improving data entry pipelines and overall SEO core web vitals.

Stack: React, Vue, Nuxt, JavaScript, Fabric.js, Canvas API, Python, GCP, GitHub Actions

Architectural Assistant — Bryden Wood

2015–2016, 2018–2021

London, UK

- Executed data-heavy computational pipelines, parametric design configurations, and complex BIM architectures. Treated physical structures as logical, highly interdependent algorithmic systems.

FEATURED PROJECT

VillageOS — Solo Full-Stack AI Engine

villageos.co.uk

- **Data Processing:** Orchestrated an LLM pipeline turning unstructured parent communications (WhatsApp messages, flyers, PDFs) into schema-valid JSON events and actionable calendar hooks.
- **Deterministic Typing:** Enforced complete full-stack safety using shared Pydantic v2 and TypeScript contracts, eliminating API boundary schema-drift bugs.
- **AI Evaluation:** Designed a gold-standard local evaluation pipeline utilising instructor to run rigorous validation benchmarks across latency, provider accuracy, and per-token spending.
- **Serverless Infrastructure:** Formed a zero-maintenance arm64 AWS Lambda setup utilising AWS SAM, protected via decoupled Supabase JWT authentication rules and PostgreSQL Row Level Security (RLS).

Stack: Next.js 16, React 19, TypeScript, FastAPI, Python, Pydantic, OpenAI SDK, Supabase (Postgres RLS), AWS Lambda, GitHub Actions (OIDC)

TECHNICAL CAPABILITIES

Languages & Frameworks

TypeScript, JavaScript, Python, Vue 3, React 19, Next.js, Nuxt, FastAPI, Node.js

Storage & Environments

PostgreSQL, Supabase (RLS), Firebase, MongoDB, Docker, AWS SAM, Lambda

AI / LLM Infrastructure

Applied LLM Engineering, Claude/MCP Servers, Instructor, OpenAI SDK, AWS Bedrock

Validation & Verification

Vitest, Playwright, Storybook, CI/CD, GitHub Actions, Tailwind CSS, Shadcn

EDUCATION

MSc & BArch Architecture

Politechnika Poznańska (2011 — 2017)
Includes Erasmus Exchange — Università di Firenze

CERTIFICATIONS & META

Claude with Amazon Bedrock

Anthropic & AWS — Issued May 2026
Languages: English (Bilingual) · Polish (Native)